



Homework 2

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Model Evaluation



Motivation

As you will recall, classification in machine learning is the problem of identifying to which of a set of categories (sub-populations) a new observation belongs, based on a training set of data containing observations (or instances) whose category membership is known.

The class an observation really belongs to is “generated” by a (physical) process / probability distribution, e.g., the height of males and females from the same background both follow Gaussian (normal) distributions.

In the last lectures, you have one specific classification method (k-nearest neighbor), but many other exist. Most of these methods have (several) parameters (e.g., k-nn has the number k and the decision criterion, or decision trees has the split strategy, the stopping criterion, etc.).

The following questions naturally arise:

- Which is the best classification method for a given problem?
- Which is the most suitable parameterization of a classification method for a given problem?

In order to answer these questions, we will be covering the following subjects:

- 1) Classifications scores
- 2) Estimation methods
- 3) Statistical tests

1) Will be covered in the homework, 2) and 3) in the lecture.



Classifications Scores

A classification score is a function that provides a quality measure for a classifier when solving a classification problem. Many such scores have been proposed, as the definition of “quality measure” may vary depending on

- What we are interested in
- What we want to optimize
- The characteristics of the problem
- The characteristics of the data set

In this homework, you will be familiarizing yourself with the following scores, which are all based on the „confusion matrix“:

1. Accuracy (Classification error)
2. Precision and Recall
3. Sensitivity and Specificity
4. F-Score

And an additional score, based on the ROC-Curve (Receiver Operating Characteristics):

1. AUC (Area under the ROC curve)



Confusion Matrix

A confusion matrix (also known as an error matrix) is a table that visualizes the performance of a classification algorithm.

- Each column of the matrix represents the instances in a predicted class
- Each row represents the instances in an actual class

The name stems from the fact that it makes it easy to see if the system is confusing two classes = mislabeling one as another.

Caveat: rows and columns are sometimes transposed and so is the order of classes!

For a binary classification problem (i.e. with two classes called “positive” and “negative”), the cells in the matrix contain the following numbers:

- **TP: true positives** = the number of observations classified as positive that are in fact positive
- **TN: true negatives** = the number of observations classified as negative that are in fact negative
- **FP: false positives** = the number of observations classified as positives that are in fact negative (also called false alarms or **type I errors**)
- **FN: false negatives** = the number of observations classified as negative that are in fact positive (also called **type II errors**)

As you can see, the correct classifications are on the diagonal.

The confusion matrix can easily be extended to multi-class classification.

		Predicted Class	
		pos	neg
Actual Class	pos	10	4
	neg	1	15

		Predicted Class	
		pos	neg
Actual Class	pos	TP	FN
	neg	FP	TN



Accuracy, Precision, Recall

Estimated time: 20min to read the Blog + 60min for the questions

We have so far mainly looked at the **accuracy**, i.e., the number of examples classified correctly by the algorithm. Using the confusion matrix, it can be defined as

$$accuracy = \frac{TP + TN}{(TP + TN + FP + FN)}$$

While this score gives a first indication of the quality of a classification algorithm for a given problem, it is not sufficient and can in fact be highly misleading!

- ♦ **Read the Blog post “[Why accuracy alone is a bad measure for classification tasks, and what we can do about it](https://tryolabs.com/blog/2013/03/25/why-accuracy-alone-bad-measure-classification-tasks-and-what-we-can-do-about-it/)”**

<https://tryolabs.com/blog/2013/03/25/why-accuracy-alone-bad-measure-classification-tasks-and-what-we-can-do-about-it/>

- ♦ **Please answer the following questions**

You are working for an insurance company on the problem of identifying fraud. You have trained two classifiers A and B and tested them on 10000 claims, the results are shown on the right.

- 1) Compute the accuracy for both models. Which one has the higher accuracy and thus, seems to be better?
- 2) Compute precision and recall for both models. The Blog posts clearly states that you want high recall for fraud detection problems. So, which model would you choose and why?
- 3) Consider your answer to the previous question once again. Do you really think it is correct? If not, why not? If yes, why?

Model A		Predicted Class	
		no fraud	fraud
Actual Class	no fraud	9700	150
	fraud	50	100

Model B		Predicted Class	
		no fraud	fraud
Actual Class	no fraud	9850	0
	fraud	100	50



Sensitivity, Specificity, F-Score

Estimated time: 30min to read the Articles + 30min for the questions

- ♦ Read the wikipedia entries “[Sensitivity and specificity](https://en.wikipedia.org/wiki/Sensitivity_and_specificity)” and “[Precision and recall](https://en.wikipedia.org/wiki/Precision_and_recall)”
https://en.wikipedia.org/wiki/Sensitivity_and_specificity
https://en.wikipedia.org/wiki/Precision_and_recall

- ♦ Please answer the following questions

- 1) What is the difference between Precision and Specificity?
Explain what having a “dog with high precision” would mean for the suitcase example!
- 2) What is the difference between Recall and Sensitivity?
- 3) Compute Sensitivity, Specificity and the F_1 -Score for the fraud Models A and B.

Model A		Predicted Class	
		no fraud	fraud
Actual Class	no fraud	9700	150
	fraud	50	100

Model B		Predicted Class	
		no fraud	fraud
Actual Class	no fraud	9850	0
	fraud	100	50



ROC and AUC

Estimated time: 14min to watch the videos + 30min for the questions

♦ Watch the Video “[ROC Curves and Area Under the Curve \(AUC\) Explained](https://www.youtube.com/watch?v=OAl6eAyP-yo)”

<https://www.youtube.com/watch?v=OAl6eAyP-yo>

♦ Please answer the following questions

- 1) Play with the ROC-Tool yourself: <http://www.navan.name/roc/>
- 2) Consider the ROC Curves in Figure 2: which model is the best, which one the worst and why? (note that this is an easy question)
- 3) Consider the ROC Curves in Figure 3, which are much more like the ones you will see in practice: which model would you choose any why? (note that this is a hard question)

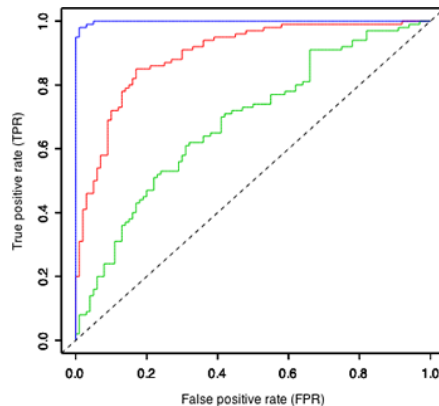


Figure 2

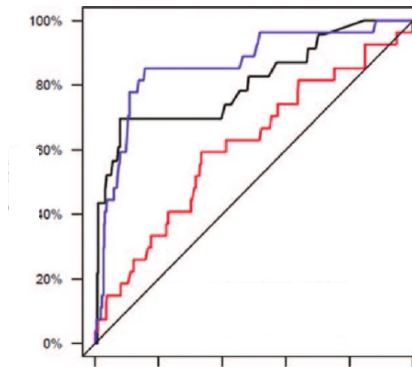


Figure 3

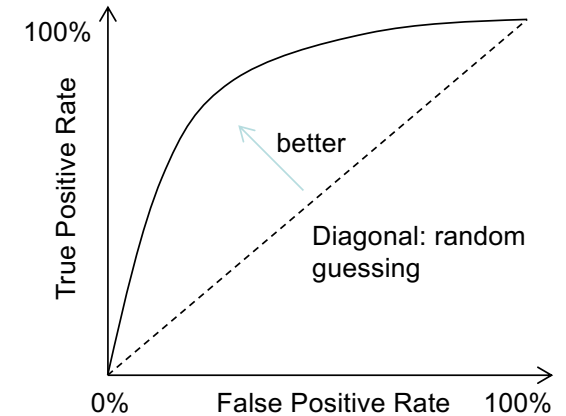


Figure 1 – The ROC Curve



Recap – Linear Algebra

You should already be familiar with linear Algebra from your Bachelor studies. In case you need a refresher, I would highly recommend the following playlist (you will not need everything immediately, so you can space it out over the next 2-3 weeks, but keep in mind that you will get additional homework every week).

Estimated time: 70min to watch the recommended videos + whatever you decide to watch

Playlist “The Essence of linear Algebra”

https://www.youtube.com/watch?v=kjBOesZCoqc&list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

This playlist consist of 11 basic chapters and a couple of footnotes, for a total 16 videos. Decide for yourself which videos you need to watch to recapitulate anything you may have forgotten.

I recommend this specific playlist as it is very graphical and non-formal, always putting the basic ideas first (that’s why it’s called the “Essence”), so it should be enjoyable to watch and easy to follow! I would recommend the watch the full playlist, but if you are firm in your math and want to skip some chapters, here are the most important ones:

- ♦ Chapters 1, 2, 3 and 4 cover basis vectors and matrix multiplications. (*total time to watch: 40min*)
- ♦ You will need to know what the determinant (chapter 5) signifies and how to compute it. (*time to watch: 10min*)
- ♦ Furthermore, we will be using matrix inversion (chapter 6) and eigenvectors (chapter 10). (*total time to watch: 30min*)



Congratulations!

You are now familiar with the confusion matrix and the most important classification scores, Accuracy (Classification error), Precision and Recall, Sensitivity and Specificity and the F-Score. In addition, you understand the ROC-Curve and the AUC score.

In the lecture, we will be looking at how to compute these scores and how to compare classifiers using the scores.